15_Examining_Missingness

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T2) Yes significant . . . 40

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0) Overall test: did drop out across the whole project?

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)              -0.934      0.252    -3.71  0.000205
## 2 Training_ConditionTRAIN   0.0808     0.354     0.228 0.819
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic       p.value
##   <chr>                      <dbl>     <dbl>     <dbl>         <dbl>
## 1 (Intercept)               -2.32      0.396    -5.85  0.00000000497
## 2 Training_ConditionTRAIN   -0.587     0.649    -0.906 0.365
```

```
##
##     CONTROL TRAIN
##   0      71    73
##   1       7     4
```

1) Child data: Checking Missingness by Questionnaire and Time-point

Across all participants:

Time	Diary	complete	partial	missing	total
T1	A	58	57	40	155
T2	A	53	54	48	155
T3	A	49	46	60	155
T1	P	60	56	39	155
T2	P	46	61	48	155
T3	P	45	53	57	155

Now same grouped by condition:

Training_Condition	Time	Diary	complete	partial	missing	total
CONTROL	T1	A	29	32	17	78
CONTROL	T2	A	31	23	24	78
CONTROL	T3	A	23	26	29	78
CONTROL	T1	P	28	33	17	78
CONTROL	T2	P	23	31	24	78
CONTROL	T3	P	24	27	27	78
TRAIN	T1	A	29	25	23	77
TRAIN	T2	A	22	31	24	77
TRAIN	T3	A	26	20	31	77
TRAIN	T1	P	32	23	22	77
TRAIN	T2	P	23	30	24	77
TRAIN	T3	P	21	26	30	77

Table of overall missingness for each diary tp by diary type:

In theory these should be the same.. but checking anyway

T1

```
##
##                      Antisocial Prosocial
##   At least some data        115       116
##   Missing                    40        39
```

T2

```
##
##                      Antisocial Prosocial
##   At least some data        107       107
##   Missing                    48        48
```

T3

```
##
##                      Antisocial Prosocial
##   At least some data         95        98
##   Missing                    60        57
```

2) Do Baseline Variables Predict Missingness?

Do baseline screening variables (Training condition, CASI, ICU, Age, FSM eligibility, EAL status, ethnicity) predict missing data at any timepoint?

Note for all analyses: timepoint_data_missing = 1 -> data fully missing for that timepoint; timepoint_data_missing = 0 -> at least partial data present

building one big model with all covariates:

T1

1) Training condition - Missing data is not associated with training condition at any timepoint

T1 (not significant)

Note - Antisocial model always reported first

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T1_A_child_missingness,
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T1_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic    p.value
##   <chr>                      <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)               -1.28      0.274     -4.66 0.00000318
## 2 Training_ConditionTRAIN    0.424     0.370      1.15 0.252
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic    p.value
##   <chr>                      <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)               -1.28      0.274    -4.66  0.00000318
## 2 Training_ConditionTRAIN    0.361     0.373     0.970 0.332
```

T2 (not significant)

```r
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)

tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)              -0.811      0.245   -3.31   0.000948
## 2 Training_ConditionTRAIN   0.0187     0.347    0.0538 0.957
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)              -0.811      0.245   -3.31   0.000948
## 2 Training_ConditionTRAIN   0.0187     0.347    0.0538 0.957
```

```r
#exp(coef(T2_A_training_missing))           # Odds ratios
#exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ Training_Condition,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 2 x 5
##   term                     estimate std.error statistic p.value
##   <chr>                       <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)               -0.525     0.234    -2.24   0.0252
## 2 Training_ConditionTRAIN    0.130     0.330     0.394  0.694
```

```
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                     estimate std.error statistic p.value
##   <chr>                       <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)               -0.636     0.238    -2.67  0.00754
## 2 Training_ConditionTRAIN    0.187     0.334     0.561 0.575
```

2) CASI

T1 (missingness associated with higher CASI for ASB but NOT PSB)

As CASI increases, the probability of missingness for ASB increases (and probability of having data decreases).

At T1, participants with higher CASI had higher odds of missing ASB data and PSB (OR = 0.08 (ASB) & 0.08 (ASB) per one point increase in CASI. Presumably the same ppts are missing data.

Note - Antisocial model always reported first

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T1_A_child_missingness,
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T1_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)        -2.05     0.418      -4.89 0.000000984
## 2 CASI.Total.Score    0.103    0.0370      2.77 0.00556
```

```
exp(coef(T1_A_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.1291213        1.1079424
```

```
exp(confint(T1_A_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.05482381 0.2847916
## CASI.Total.Score 1.03134403 1.1936120
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)        -2.13     0.424      -5.02 0.000000530
## 2 CASI.Total.Score    0.107    0.0373      2.87 0.00415
```

```r
exp(coef(T1_P_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.1192357        1.1127280
```

```r
exp(confint(T1_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.04995047 0.2655598
## CASI.Total.Score 1.03528917 1.1995990
```

T2 (missingness associated with CASI for ASB and PSB)

At T2, participants with higher CASI had higher odds of missing ASB data and PSB (OR = 1.115 (ASB) & 1.115 (ASB) per one point increase in CASI. Presumably the same ppts are missing data.

```r
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)

tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)        -2.06     0.412      -5.01 0.000000555
## 2 CASI.Total.Score    0.131    0.0374      3.51 0.000447
```

```r
exp(coef(T2_A_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.1272421        1.1403952
```

```r
exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.05463625 0.2766772
## CASI.Total.Score 1.06203846 1.2311328
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)        -2.06     0.412      -5.01 0.000000555
## 2 CASI.Total.Score    0.131    0.0374      3.51 0.000447
```

```r
exp(coef(T2_P_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.1272421        1.1403952
```

```r
exp(confint(T2_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.05463625 0.2766772
## CASI.Total.Score 1.06203846 1.2311328
```

T3) (missingness associated with CASI for both ASB and PSB) At T3, participants with higher CASI had higher odds of missing ASB data and PSB (OR = 0.02 (ASB) & 0.04 (PSB) per one point increase in CASI.

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ CASI.Total.Score,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -1.41     0.375      -3.75 0.000174
## 2 CASI.Total.Score    0.102    0.0356      2.86 0.00427
```

```r
exp(coef(T3_A_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.2447517        1.1070741
```

```r
exp(confint(T3_A_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.1144037 0.5007296
## CASI.Total.Score 1.0342826 1.1903104
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic   p.value
##   <chr>               <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)        -1.68     0.389      -4.33 0.0000148
## 2 CASI.Total.Score    0.122    0.0367      3.32 0.000902
```

```r
exp(coef(T3_P_training_missing))           # Odds ratios
```

```
##      (Intercept) CASI.Total.Score
##        0.1855427        1.1294167
```

```r
exp(confint(T3_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.08394245 0.3881111
## CASI.Total.Score 1.05345590 1.2174485
```

3) ICU

T1 (missingness associated with ICU for ASB and PSB)

As ICU increases, the probability of missingness for both ASB and PSB increases (and probability of having data decreases)

At T1, participants with higher ICU had higher odds of missing ASB data and PSB (OR = 1.03 (ASB) & 1.03 (ASB) per one point increase in ICU Presumably the same ppts are missing data.

Note - Antisocial model always reported first

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T1_A_child_missingness,
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T1_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  -2.48      0.830      -2.99 0.00279
## 2 ICU.Score     0.0307    0.0171      1.79 0.0728
```

```
exp(coef(T1_A_training_missing))           # Odds ratios
```

```
## (Intercept)   ICU.Score
##  0.08359332  1.03115180
```

```
exp(confint(T1_A_training_missing))        # 95% CIs for odds ratios
```

```
##                  2.5 %   97.5 %
## (Intercept) 0.01540883 0.405975
## ICU.Score   0.99752951 1.067076
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  -2.47      0.835      -2.95 0.00316
## 2 ICU.Score     0.0296    0.0172      1.72 0.0854
```

```r
exp(coef(T1_P_training_missing))           # Odds ratios
```

```
## (Intercept)   ICU.Score
##  0.08498912  1.03003570
```

```r
exp(confint(T1_P_training_missing))        # 95% CIs for odds ratios
```

```
##                  2.5 %   97.5 %
## (Intercept) 0.01549995 0.416909
## ICU.Score   0.99620956 1.066131
```

T2 (missingness associated with ICU for PSB and ASB)

At T2, participants with higher ICU had higher odds of missing ASB data and PSB (OR = 1.04 (ASB) & 1.04 (ASB) per one point increase in ICU Presumably the same ppts are missing data.

```r
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  -2.52      0.794      -3.17 0.00150
## 2 ICU.Score     0.0370    0.0164      2.26 0.0240
```

```r
exp(coef(T2_A_training_missing))           # Odds ratios
```

```
## (Intercept)   ICU.Score
##  0.08033915  1.03774448
```

```r
exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```
##                  2.5 %    97.5 %
## (Intercept) 0.01595646 0.3647857
## ICU.Score   1.00539923 1.0725516
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  -2.52      0.794      -3.17 0.00150
## 2 ICU.Score     0.0370    0.0164      2.26 0.0240
```

```r
exp(coef(T2_P_training_missing))           # Odds ratios
```

```
## (Intercept)   ICU.Score
##  0.08033915  1.03774448
```

```r
exp(confint(T2_P_training_missing))        # 95% CIs for odds ratios
```

```
##                  2.5 %    97.5 %
## (Intercept) 0.01595646 0.3647857
## ICU.Score   1.00539923 1.0725516
```

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ ICU.Score,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)  -1.37      0.722      -1.90  0.0578
## 2 ICU.Score     0.0199    0.0152      1.30  0.193
```

4) Age

T1 (missingness associated with age for both ASB and PSB)

As Age increases, the probability of missingness for both ASB and PSB increases (and probability of having data decreases)

Older participants had higher odds of missing ASB data and PSB at T1 (OR = 1.66 (ASB) & 1.77 (ASB)) per one year increase in age.

Note - Antisocial model always reported first

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T1_A_child_missingness,
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T1_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -8.91      2.56      -3.48 0.000507
## 2 Age..Screening.    0.566     0.182      3.11 0.00189
```

```
exp(coef(T1_A_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##    0.0001343979    1.7618725673
```

```
exp(confint(T1_A_training_missing))        # 95% CIs for odds ratios
```

```
##                        2.5 %     97.5 %
## (Intercept)     6.992842e-07 0.01710653
## Age..Screening. 1.245603e+00 2.55518441
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -9.45      2.62      -3.61 0.000304
## 2 Age..Screening.    0.602     0.186      3.24 0.00119
```

```r
exp(coef(T1_P_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##    7.834485e-05    1.826161e+00
```

```r
exp(confint(T1_P_training_missing))        # 95% CIs for odds ratios
```

```
##                        2.5 %    97.5 %
## (Intercept)     3.602778e-07 0.0109077
## Age..Screening. 1.283544e+00 2.6696958
```

T2 (missingness associated with age for PSB and ASB)

Older participants had higher odds of missing ASB data and PSB at T2 (OR = 1.64 (ASB) & 1.64 (PSB)) per one year increase in age.

```r
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -8.27      2.40      -3.45 0.000564
## 2 Age..Screening.    0.540     0.171      3.15 0.00164
```

```r
exp(coef(T2_A_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##    0.0002567234    1.7153668495
```

```r
exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```
##                        2.5 %     97.5 %
## (Intercept)     1.925187e-06 0.02430473
## Age..Screening. 1.237114e+00 2.43008638
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -8.27      2.40      -3.45 0.000564
## 2 Age..Screening.    0.540     0.171      3.15 0.00164
```

```r
exp(coef(T2_P_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##    0.0002567234    1.7153668495
```

```r
exp(confint(T2_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %     97.5 %
## (Intercept)     1.925187e-06 0.02430473
## Age..Screening. 1.237114e+00 2.43008638
```

T3) (missingness associated with age for both PSB and ASB) Older participants had higher odds of missing ASB data (OR = 1.52) and PSB (OR = 1.67) at T3 per one year increase in age.

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ Age..Screening.,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -6.48      2.20      -2.95 0.00322
## 2 Age..Screening.    0.437     0.159      2.76 0.00585
```

```r
exp(coef(T3_A_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##     0.001533882     1.548224779
```

```r
exp(confint(T3_A_training_missing))        # 95% CIs for odds ratios
```

```
##                         2.5 %    97.5 %
## (Intercept)     1.789936e-05 0.1031868
## Age..Screening. 1.142073e+00 2.1323047
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)       -8.14      2.30      -3.54 0.000399
## 2 Age..Screening.    0.550     0.165      3.33 0.000859
```

```r
exp(coef(T3_P_training_missing))           # Odds ratios
```

```
##     (Intercept) Age..Screening.
##    0.0002928234    1.7337684986
```

```r
exp(confint(T3_P_training_missing))        # 95% CIs for odds ratios
```

```
##                         2.5 %     97.5 %
## (Intercept)     2.718885e-06 0.02310333
## Age..Screening. 1.265529e+00 2.42431422
```

5) FSM eligibility

(FSM 1 = yes, 2 = no, 3 = IDK, NA/4 = missing) **here I have removed all participants who were labelled 'unsure' as it confuses results and generates weird output.**

T1 (missingness associated with FSM for ASB and PSB)

- ASB: Participants with FSM were 0.55 x more likely to have missing ASB data at T1 than those without.
- PSB: Participants with FSM were 0.36 x more likely to have missing PSB data at T1 than those without.

Note - Antisocial model always reported first

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_A_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_P_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250    -3.87  0.000108
## 2 FSM_eligibility2   -0.352     0.410    -0.859 0.390
```

```
exp(coef(T1_A_training_missing))           # Odds ratios
```

```
##      (Intercept) FSM_eligibility2
##        0.3793103        0.7030303
```

```
exp(confint(T1_A_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.2274167 0.6100648
## FSM_eligibility2 0.3072614 1.5510194
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250     -3.87 0.000108
## 2 FSM_eligibility2   -0.461     0.419     -1.10 0.271
```

```
exp(coef(T1_P_training_missing))           # Odds ratios
```

```
##      (Intercept) FSM_eligibility2
##        0.3793103        0.6304348
```

```
exp(confint(T1_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.2274167 0.6100648
## FSM_eligibility2 0.2696040 1.4093477
```

T2 (missingness associated with FSM for PSB and ASB)

- ASB: Participants with FSM were 0.52 x more likely to have missing ASB data at T2 than those without.
- PSB: Participants with FSM were 0.52 x more likely to have missing PSB data at T2 than those without.

```
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_A_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_P_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)



tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250    -3.87  0.000108
## 2 FSM_eligibility2   -0.352     0.410    -0.859 0.390
```

```
exp(coef(T2_A_training_missing))           # Odds ratios
```

```
##      (Intercept) FSM_eligibility2
##        0.3793103        0.7030303
```

```r
exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.2274167 0.6100648
## FSM_eligibility2 0.3072614 1.5510194
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250     -3.87 0.000108
## 2 FSM_eligibility2   -0.461     0.419     -1.10 0.271
```

```r
exp(coef(T2_P_training_missing))           # Odds ratios
```

```
##      (Intercept) FSM_eligibility2
##        0.3793103        0.6304348
```

```r
exp(confint(T2_P_training_missing))        # 95% CIs for odds ratios
```

```
##                       2.5 %    97.5 %
## (Intercept)      0.2274167 0.6100648
## FSM_eligibility2 0.2696040 1.4093477
```

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_A_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ FSM_eligibility,
  data = subset(T1_A_child_missingness, FSM_eligibility == 1 | FSM_eligibility == 2),
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250    -3.87  0.000108
## 2 FSM_eligibility2   -0.352     0.410    -0.859 0.390
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic  p.value
##   <chr>               <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        -0.969     0.250    -3.87  0.000108
## 2 FSM_eligibility2   -0.352     0.410    -0.859 0.390
```

6) EAL

Has EAL - 1 = yes, 2 = no, 3 IDK **again, removing unsure responses**

Note - Antisocial model always reported first

T1 (not significant)

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
  data = subset(T1_A_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
  data = subset(T1_P_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.397     0.480    -0.827  0.408
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.448     0.482    -0.929  0.353
```

T2 (not significant)

```
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
  data = subset(T1_A_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
```

```
  data = subset(T1_A_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)



tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.397     0.480    -0.827  0.408
```

```
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.397     0.480    -0.827  0.408
```

```
#exp(coef(T2_A_training_missing))           # Odds ratios
#exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```


# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
  data = subset(T1_A_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ has_EAL,
  data = subset(T1_A_child_missingness, has_EAL == 1 | has_EAL == 2),
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.397     0.480    -0.827  0.408
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   -0.811     0.425    -1.91   0.0563
## 2 has_EAL2      -0.397     0.480    -0.827  0.408
```

7) Ethnicity

T1 (not significant)

Note - Antisocial model always reported first

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -2.56       1.04   -2.47    0.0134
## 2 cEthnicityBlack  -16.0     1496.     -0.0107  0.991
## 3 cEthnicityMixed   -0.208      1.46   -0.142   0.887
## 4 cEthnicityOther    1.06       1.30    0.817   0.414
## 5 cEthnicityWhite    0.296      1.12    0.264   0.792
```

```
## # A tibble: 5 x 5
##   term             estimate std.error statistic p.value
##   <chr>               <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)     -1.96e+ 1     2874. -6.81e- 3   0.995
## 2 cEthnicityBlack  4.02e-11     3788.  1.06e-14   1.000
## 3 cEthnicityMixed  1.68e+ 1     2874.  5.84e- 3   0.995
## 4 cEthnicityOther  1.81e+ 1     2874.  6.28e- 3   0.995
## 5 cEthnicityWhite  1.73e+ 1     2874.  6.02e- 3   0.995
```

T2 (not significant)

```
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ cEthnicity,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
  timepoint_data_missing ~ cEthnicity,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T2_A_training_missing)
```

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -1.30      0.651    -1.99    0.0461
## 2 cEthnicityBlack   -1.59      1.22     -1.31    0.191
## 3 cEthnicityMixed   -1.47      1.22     -1.21    0.227
## 4 cEthnicityOther   -0.205     1.02     -0.201   0.840
## 5 cEthnicityWhite   -0.273     0.731    -0.374   0.709
```

```r
tidy(T2_P_training_missing)
```

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -1.30      0.651    -1.99   0.0461
## 2 cEthnicityBlack   -1.59      1.22     -1.31   0.191
## 3 cEthnicityMixed   -1.47      1.22     -1.21   0.227
## 4 cEthnicityOther   -0.205     1.02     -0.201  0.840
## 5 cEthnicityWhite   -0.273     0.731    -0.374  0.709
```

```r
#exp(coef(T2_A_training_missing))           # Odds ratios
#exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ cEthnicity,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ cEthnicity,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -0.288     0.540    -0.533  0.594
## 2 cEthnicityBlack   -1.39      0.829    -1.67   0.0945
## 3 cEthnicityMixed   -0.891     0.787    -1.13   0.257
## 4 cEthnicityOther   -2.01      1.18     -1.71   0.0876
## 5 cEthnicityWhite   -0.811     0.612    -1.32   0.185
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)       -0.916     0.592    -1.55    0.121
## 2 cEthnicityBlack   -1.22      0.953    -1.28    0.199
```

```
## 3 cEthnicityMixed   -0.262     0.823    -0.319   0.750
## 4 cEthnicityOther   -1.39      1.20     -1.15    0.250
## 5 cEthnicityWhite   -0.182     0.658    -0.277   0.782
```

7) School Type

Note - Antisocial model always reported first

T1 (not significant)

```
# T1
T1_A_training_missing <- glm(
  timepoint_data_missing ~ school_type,
  data = T1_A_child_missingness,
  family = binomial(link = "logit")
)

T1_P_training_missing <- glm(
  timepoint_data_missing ~ school_type,
  data = T1_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T1_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              -19.6     2195.  -0.00891   0.993
## 2 school_typeMainstream     17.4     2195.   0.00791   0.994
```

```
tidy(T1_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              -19.6     2195.  -0.00891   0.993
## 2 school_typeMainstream     17.2     2195.   0.00785   0.994
```

T2 (missingness associated with school type)

At T2, participants in mainstream schools FSM were 0.23 x more likely to have missing data than those at AP schools.

```
# T2
T2_A_training_missing <- glm(
  timepoint_data_missing ~ school_type,
  data = T2_A_child_missingness,
  family = binomial(link = "logit")
)

T2_P_training_missing <- glm(
```

```
  timepoint_data_missing ~ school_type,
  data = T2_P_child_missingness,
  family = binomial(link = "logit")
)



tidy(T2_A_training_missing)
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             -0.693     0.433     -1.60 0.109
## 2 school_typeMainstream   -1.52      0.546     -2.77 0.00555
```

```
exp(coef(T2_A_training_missing))           # Odds ratios
```

```
##           (Intercept) school_typeMainstream
##             0.5000000             0.2197802
```

```
exp(confint(T2_A_training_missing))        # 95% CIs for odds ratios
```

```
##                            2.5 %    97.5 %
## (Intercept)           0.20272874 1.1366490
## school_typeMainstream 0.07474471 0.6524142
```

```
tidy(T2_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             -0.693     0.433     -1.60 0.109
## 2 school_typeMainstream   -1.52      0.546     -2.77 0.00555
```

```
exp(coef(T2_P_training_missing))           # Odds ratios
```

```
##           (Intercept) school_typeMainstream
##             0.5000000             0.2197802
```

```
exp(confint(T2_P_training_missing))        # 95% CIs for odds ratios
```

```
##                            2.5 %    97.5 %
## (Intercept)           0.20272874 1.1366490
## school_typeMainstream 0.07474471 0.6524142
```

```r
# T2
T3_A_training_missing <- glm(
  timepoint_data_missing ~ school_type,
  data = T3_A_child_missingness,
  family = binomial(link = "logit")
)

T3_P_training_missing <- glm(
  timepoint_data_missing ~ school_type,
  data = T3_P_child_missingness,
  family = binomial(link = "logit")
)


tidy(T3_A_training_missing)
```

T3) (not significant)

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             -0.887     0.449    -1.98   0.0482
## 2 school_typeMainstream   -0.334     0.508    -0.657  0.511
```

```r
tidy(T3_P_training_missing)
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)             -0.887     0.449    -1.98   0.0482
## 2 school_typeMainstream   -0.511     0.514    -0.995  0.320
```

Teacher data

1) Age

T1) Yes significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic   p.value
##   <chr>              <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)       11.3       2.76       4.07 0.0000464
## 2 Age..Screening.   -0.728     0.195     -3.73 0.000194
```

```
##     (Intercept) Age..Screening.
##    7.775530e+04    4.829681e-01
```

```
##                       2.5 %       97.5 %
## (Intercept)     439.7975723 2.384619e+07
## Age..Screening.   0.3230548 6.977361e-01
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic   p.value
##   <chr>              <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)       11.3       2.76       4.07 0.0000464
## 2 Age..Screening.   -0.728     0.195     -3.73 0.000194
```

```
##     (Intercept) Age..Screening.
##    7.775530e+04    4.829681e-01
```

```
##                      2.5 %       97.5 %
## (Intercept)     439.7975723 2.384619e+07
## Age..Screening.   0.3230548 6.977361e-01
```

T2) Yes significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        8.27      2.40       3.45 0.000564
## 2 Age..Screening.   -0.540     0.171     -3.15 0.00164
```

```
##     (Intercept) Age..Screening.
##    3895.2428186       0.5829657
```

```
##                     2.5 %       97.5 %
## (Intercept)     41.144262 5.194302e+05
## Age..Screening.  0.411508 8.083330e-01
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        8.27      2.40       3.45 0.000564
## 2 Age..Screening.   -0.540     0.171     -3.15 0.00164
```

```
##     (Intercept) Age..Screening.
##    3895.2428186       0.5829657
```

```
##                     2.5 %       97.5 %
## (Intercept)     41.144262 5.194302e+05
## Age..Screening.  0.411508 8.083330e-01
```

T3) Yes significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        7.87      2.32       3.39 0.000693
## 2 Age..Screening.   -0.522     0.166     -3.14 0.00169
```

```
##     (Intercept) Age..Screening.
##    2616.9049663       0.5931036
```

```
##                       2.5 %       97.5 %
## (Intercept)     31.7312581 2.942665e+05
## Age..Screening.  0.4231985 8.148856e-01
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic  p.value
##   <chr>              <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)        7.87      2.32       3.39 0.000693
## 2 Age..Screening.   -0.522     0.166     -3.14 0.00169
```

```
##     (Intercept) Age..Screening.
##    2616.9049663       0.5931036
```

```
##                       2.5 %       97.5 %
## (Intercept)     31.7312581 2.942665e+05
## Age..Screening.  0.4231985 8.148856e-01
```

2) CASI

T1) Yes significant

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)         2.15     0.427       5.04 0.000000471
## 2 CASI.Total.Score   -0.106    0.0374     -2.83 0.00472
```

```
##      (Intercept) CASI.Total.Score
##        8.5993103        0.8997193
```

```
##                       2.5 %    97.5 %
## (Intercept)      3.8397283 20.663907
## CASI.Total.Score 0.8344016  0.967378
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)         2.15     0.427       5.04 0.000000471
## 2 CASI.Total.Score   -0.106    0.0374     -2.83 0.00472
```

```
##      (Intercept) CASI.Total.Score
##        8.5993103        0.8997193
```

```
##                       2.5 %    97.5 %
## (Intercept)      3.8397283 20.663907
## CASI.Total.Score 0.8344016  0.967378
```

T2) Yes significant

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)         2.06     0.412       5.01 0.000000555
## 2 CASI.Total.Score   -0.131    0.0374     -3.51 0.000447
```

```
##      (Intercept) CASI.Total.Score
##         7.859032         0.876889
```

```
##                      2.5 %     97.5 %
## (Intercept)      3.6143200 18.3028668
## CASI.Total.Score 0.8122601  0.9415855
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic     p.value
##   <chr>               <dbl>     <dbl>     <dbl>       <dbl>
## 1 (Intercept)         2.06     0.412       5.01 0.000000555
## 2 CASI.Total.Score   -0.131    0.0374     -3.51 0.000447
```

```
##      (Intercept) CASI.Total.Score
##         7.859032         0.876889
```

```
##                      2.5 %     97.5 %
## (Intercept)      3.6143200 18.3028668
## CASI.Total.Score 0.8122601  0.9415855
```

T3) Yes significant

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic   p.value
##   <chr>               <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)         1.68     0.389       4.33 0.0000150
## 2 CASI.Total.Score   -0.109    0.0360     -3.02 0.00253
```

```
##      (Intercept) CASI.Total.Score
##        5.3810839        0.8968725
```

```
##                      2.5 %     97.5 %
## (Intercept)      2.5706828 11.8796245
## CASI.Total.Score 0.8334697  0.9609063
```

```
## # A tibble: 2 x 5
##   term             estimate std.error statistic   p.value
##   <chr>               <dbl>     <dbl>     <dbl>     <dbl>
## 1 (Intercept)         1.68     0.389       4.33 0.0000150
## 2 CASI.Total.Score   -0.109    0.0360     -3.02 0.00253
```

```
##      (Intercept) CASI.Total.Score
##        5.3810839        0.8968725
```

```
##                      2.5 %     97.5 %
## (Intercept)      2.5706828 11.8796245
## CASI.Total.Score 0.8334697  0.9609063
```

3) ICU

T1) Not significant

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   2.42      0.840      2.88 0.00395
## 2 ICU.Score    -0.0279    0.0173    -1.61 0.107
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   2.42      0.840      2.88 0.00395
## 2 ICU.Score    -0.0279    0.0173    -1.61 0.107
```

T2) Yes significant

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   2.52      0.794      3.17 0.00150
## 2 ICU.Score    -0.0370    0.0164    -2.26 0.0240
```

```
## (Intercept)   ICU.Score
##  12.4472317   0.9636283
```

```
##                 2.5 %     97.5 %
## (Intercept) 2.7413359 62.6705257
## ICU.Score   0.9323561  0.9946298
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   2.52      0.794      3.17 0.00150
## 2 ICU.Score    -0.0370    0.0164    -2.26 0.0240
```

```
## (Intercept)   ICU.Score
##  12.4472317   0.9636283
```

```
##                 2.5 %     97.5 %
## (Intercept) 2.7413359 62.6705257
## ICU.Score   0.9323561  0.9946298
```

T3) Not significant

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   1.50      0.742      2.02  0.0438
## 2 ICU.Score    -0.0183    0.0156    -1.17  0.241
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   1.50      0.742      2.02  0.0438
## 2 ICU.Score    -0.0183    0.0156    -1.17  0.241
```

4) FSM Eligibility

T1) Not significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.962     0.524     1.84   0.0664
## 2 FSM_eligibility    0.154     0.328     0.469  0.639
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.962     0.524     1.84   0.0664
## 2 FSM_eligibility    0.154     0.328     0.469  0.639
```

T2) NOT significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.215     0.488     0.441   0.659
## 2 FSM_eligibility    0.422     0.313     1.35    0.177
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.215     0.488     0.441   0.659
## 2 FSM_eligibility    0.422     0.313     1.35    0.177
```

T3) Not significant

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.152     0.473     0.322   0.748
## 2 FSM_eligibility    0.356     0.299     1.19    0.234
```

```
## # A tibble: 2 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.152     0.473     0.322   0.748
## 2 FSM_eligibility    0.356     0.299     1.19    0.234
```

5) Has EAL

T1) Not significant

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic    p.value
##   <chr>                      <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)                1.20      0.269     4.48  0.00000746
## 2 Training_ConditionTRAIN   -0.157     0.374    -0.419 0.675
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)    0.664     0.802     0.827   0.408
## 2 has_EAL        0.285     0.423     0.673   0.501
```

T2) NOT significant

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   0.761      0.757     1.01    0.314
## 2 has_EAL       0.0426     0.394     0.108   0.914
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)   0.761      0.757     1.01    0.314
## 2 has_EAL       0.0426     0.394     0.108   0.914
```

T3) Not significant

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)    0.241     0.729     0.331   0.741
## 2 has_EAL        0.237     0.382     0.621   0.534
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic p.value
##   <chr>          <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)    0.241     0.729     0.331   0.741
## 2 has_EAL        0.237     0.382     0.621   0.534
```

6) Training Condition

T1) Not significant

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic    p.value
##   <chr>                      <dbl>     <dbl>     <dbl>      <dbl>
```

```
## 1 (Intercept)                1.20      0.269     4.48  0.00000746
## 2 Training_ConditionTRAIN   -0.157     0.374    -0.419 0.675
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic    p.value
##   <chr>                      <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)                1.20      0.269     4.48  0.00000746
## 2 Training_ConditionTRAIN   -0.157     0.374    -0.419 0.675
```

T2) NOT significant

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)                0.811     0.245     3.31   0.000948
## 2 Training_ConditionTRAIN   -0.0187    0.347    -0.0538 0.957
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)                0.811     0.245     3.31   0.000948
## 2 Training_ConditionTRAIN   -0.0187    0.347    -0.0538 0.957
```

T3) Not significant

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic p.value
##   <chr>                      <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)                0.751     0.243     3.10  0.00195
## 2 Training_ConditionTRAIN   -0.192     0.339    -0.566 0.572
```

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic p.value
##   <chr>                      <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)                0.751     0.243     3.10  0.00195
## 2 Training_ConditionTRAIN   -0.192     0.339    -0.566 0.572
```

7) Ethnicity

T1) Not significant

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)         18.6     1743.   0.0107    0.992
## 2 cEthnicityBlack    -15.7     1743.  -0.00899   0.993
## 3 cEthnicityMixed    -15.8     1743.  -0.00906   0.993
## 4 cEthnicityOther    -17.1     1743.  -0.00979   0.992
## 5 cEthnicityWhite    -16.1     1743.  -0.00923   0.993
```

```
## # A tibble: 2 x 5
##   term                     estimate std.error statistic    p.value
##   <chr>                       <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)                 1.20      0.269     4.48  0.00000746
## 2 Training_ConditionTRAIN    -0.157     0.374    -0.419 0.675
```

T2) NOT significant

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        1.30      0.651     1.99   0.0461
## 2 cEthnicityBlack    1.59      1.22      1.31   0.191
## 3 cEthnicityMixed    1.47      1.22      1.21   0.227
## 4 cEthnicityOther    0.205     1.02      0.201  0.840
## 5 cEthnicityWhite    0.273     0.731     0.374  0.709
```

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        1.30      0.651     1.99   0.0461
## 2 cEthnicityBlack    1.59      1.22      1.31   0.191
## 3 cEthnicityMixed    1.47      1.22      1.21   0.227
## 4 cEthnicityOther    0.205     1.02      0.201  0.840
## 5 cEthnicityWhite    0.273     0.731     0.374  0.709
```

T3) Not significant

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.916     0.592     1.55    0.121
## 2 cEthnicityBlack    1.22      0.953     1.28    0.199
## 3 cEthnicityMixed    0.624     0.869     0.718   0.472
## 4 cEthnicityOther    1.39      1.20      1.15    0.250
## 5 cEthnicityWhite    0.451     0.668     0.674   0.500
```

```
##     (Intercept) cEthnicityBlack cEthnicityMixed cEthnicityOther cEthnicityWhite
##        2.500000        3.400000        1.866667        4.000000        1.569231
```

```
##                     2.5 %    97.5 %
## (Intercept)     0.8362395  9.114195
## cEthnicityBlack 0.5590071 27.900073
## cEthnicityMixed 0.3390636 11.322630
## cEthnicityOther 0.4826001 85.894757
## cEthnicityWhite 0.3827103  5.574740
```

```
## # A tibble: 5 x 5
##   term            estimate std.error statistic p.value
##   <chr>              <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)        0.916     0.592     1.55    0.121
## 2 cEthnicityBlack    1.22      0.953     1.28    0.199
```

```
## 3 cEthnicityMixed    0.624     0.869     0.718   0.472
## 4 cEthnicityOther    1.39      1.20      1.15    0.250
## 5 cEthnicityWhite    0.451     0.668     0.674   0.500
```

```
##     (Intercept) cEthnicityBlack cEthnicityMixed cEthnicityOther cEthnicityWhite
##        2.500000        3.400000        1.866667        4.000000        1.569231
```

```
##                     2.5 %    97.5 %
## (Intercept)     0.8362395  9.114195
## cEthnicityBlack 0.5590071 27.900073
## cEthnicityMixed 0.3390636 11.322630
## cEthnicityOther 0.4826001 85.894757
## cEthnicityWhite 0.3827103  5.574740
```

8) School Type

T1) Not significant

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)               19.6     2195.   0.00891   0.993
## 2 school_typeMainstream    -17.2     2195.  -0.00785   0.994
```

```
## # A tibble: 2 x 5
##   term                     estimate std.error statistic    p.value
##   <chr>                       <dbl>     <dbl>     <dbl>      <dbl>
## 1 (Intercept)                 1.20      0.269     4.48  0.00000746
## 2 Training_ConditionTRAIN    -0.157     0.374    -0.419 0.675
```

T2) Yes significant

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              0.693     0.433      1.60 0.109
## 2 school_typeMainstream    1.52      0.546      2.77 0.00555
```

```
##           (Intercept) school_typeMainstream
##                  2.00                  4.55
```

```
##                           2.5 %   97.5 %
## (Intercept)           0.8797791  4.93270
## school_typeMainstream 1.5327686 13.37887
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              0.693     0.433      1.60 0.109
## 2 school_typeMainstream    1.52      0.546      2.77 0.00555
```

```
##           (Intercept) school_typeMainstream
##                  2.00                  4.55
```

```
##                           2.5 %   97.5 %
## (Intercept)           0.8797791  4.93270
## school_typeMainstream 1.5327686 13.37887
```

T3) Yes significant

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              0.693     0.433      1.60  0.109
## 2 school_typeMainstream    1.05      0.516      2.04  0.0411
```

```
##           (Intercept) school_typeMainstream
##              2.000000              2.866667
```

```
##                           2.5 %   97.5 %
## (Intercept)           0.8797791 4.932700
## school_typeMainstream 1.0129278 7.812131
```

```
## # A tibble: 2 x 5
##   term                  estimate std.error statistic p.value
##   <chr>                    <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)              0.693     0.433      1.60  0.109
## 2 school_typeMainstream    1.05      0.516      2.04  0.0411
```

```
##           (Intercept) school_typeMainstream
##              2.000000              2.866667
```

```
##                           2.5 %   97.5 %
## (Intercept)           0.8797791 4.932700
## school_typeMainstream 1.0129278 7.812131
```

3) Predicting consent by IBT condition

First: All children that have an IQ of '0' either rejected the project or were not at school for S1. i.e. they are missing baseline data AND they are down as not consenting so we have confirmation from two sources. Demonstrating this:

```
## # A tibble: 9 x 3
## # Groups:   dropped_out_reason [9]
##   dropped_out_reason           missing_T0     n
##   <fct>                             <dbl> <int>
## 1 -                                     0   110
## 2 Refused                               1    11
## 3 Dropped out during S1                 1     3
## 4 Dropped out before training           0     6
## 5 Dropped out during training           0     5
```

```
## 6 Dropped out before/during D1          0     1
## 7 Dropped out before/during D2          0     2
## 8 Dropped out before/during D3          0     1
## 9 Nonattender/noshow                    1    16
```

Now testing whether this varies by IBT condition

```
## # A tibble: 2 x 5
##   term                    estimate std.error statistic  p.value
##   <chr>                      <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)               0.179     0.0450     3.99  0.000103
## 2 Training_ConditionTRAIN   0.0283    0.0638     0.443 0.658
```